

Name: _____

Date: _____

Class: _____

MyDesign Portfolio

Element I Initial Template

Test 1:

Our first test is a high priority and involves measuring the height of our prototype. First, have a tape measurer and the prototype. Next, put the tape measurer on the top end of the stick and run the tape measure all the way down to the bottom. Finally, record the height of the stick from the tape measurer. Pass: The height is in the expected range of 12-15 feet in height or is above the range. Fail: The height is under the expected range of 12-15 feet. Goal: Have the prototype to be in the range of 12-15 feet as that should be enough for the prototype to work. We also have an extension piece which could be used to reach the height.

Result: The prototype was 13ft and 2 inches long, which means the prototype passes the requirement. The data shows that the prototype is longer than 12 ft and can reach the badminton birdies.

Test 2: The second test is a high priority and involves the stability of the prototype. It must be structurally straight and stable. First, have the stick prototype held in an open space horizontally and parallel to the ground. Next, record if there is any bends or weak spots in the design. Closely examine where the segments connect as there could be weak spots. Finally, record if the stick prototype has a bend angle or not. Pass: The stick is either perfectly straight or has few spots of weakness or bends. Fail: The prototype has several bends and isn't stable. Goal: To have our prototype be straight in shape as it will be the most stable when held vertically.

Result: Our prototype had minimal bends and weak points, which means the prototype passes the requirement. The trends from the data conclude that the prototype is at a suitable and proper angle, and can be structurally sound vertically.

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Test 3: The third and final test is a Mobility/Transportation test where the prototype must be transported 20 feet and reassembled within 2 minutes. First, have the prototype fully together. Next, disassemble the prototype into the 4 segments. Take the segments and move them to their final destination. Finally, put the prototype back together. Pass: The prototype can be disassembled, transported 20 feet, and constructed successfully or with minimal errors within the time frame. Fail: The prototype can't be transported, put back together, is not secure, or can't be disassembled within the time frame. Goal: To have the prototype be transported from A to B in a timely manner without problems. This is a key component of our design and it is important for the stakeholders.

Result: The prototype has been taken apart, moved, and put back together numerous times, which means that the prototype passes the test. The data shows that the prototype exceeded our expectations with it being moved in under 2 minutes.

Data Conclusion: Our prototype passes all of the testing requirements as it reaches over 12ft, can be structurally sound, and can be disassembled, moved, and put back together in a reasonable amount of time.

Element I: Testing, Data Collection and Analysis						
	5	4	3	2	1	0

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1. Quantity of tests conducted. The student conducts _____ test s for high priority requirements that are reasonable based on instructional context, or through physical or mathematical modeling.	<i>more than four</i>	<i>four</i>	<i>three</i>	<i>two</i>	<i>one</i>	<i>no</i>
2. Importance of tests. The student conducts tests for _____ requirements that are reasonable based on instructional context, or through physical or mathematical modeling.	<i>all high priority</i>	<i>most high priority</i>	<i>half of high priority</i>	<i>some low and high priority</i>	<i>only non priority</i>	<i>no</i>
3. Comprehension of testing procedure. The student demonstrates _____ understanding of testing procedure, including the gathering and analysis of resultant data.	<i>considerable</i>	<i>ample</i>	<i>adequate</i>	<i>partial or overly general</i>	<i>minimal</i>	<i>a failure of</i>

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4. Detail of the design analysis. The analysis of the effectiveness with which the design met stated goals includes _____ explanation (and summary) of the data.	<i>a consistently detailed (generously supported by pictures, graphs, charts and other visuals)</i>	<i>a generally detailed (generally supported by pictures, graphs, charts and other visuals)</i>	<i>a somewhat detailed (at least somewhat supported by pictures, graphs, charts and other visuals)</i>	<i>a partial (partially complete and/or partially correct and minimally supported by pictures, graphs, and other visuals)</i>	<i>an attempted (may not be supported by pictures, graphs, charts and other visuals)</i>	<i>no</i>
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